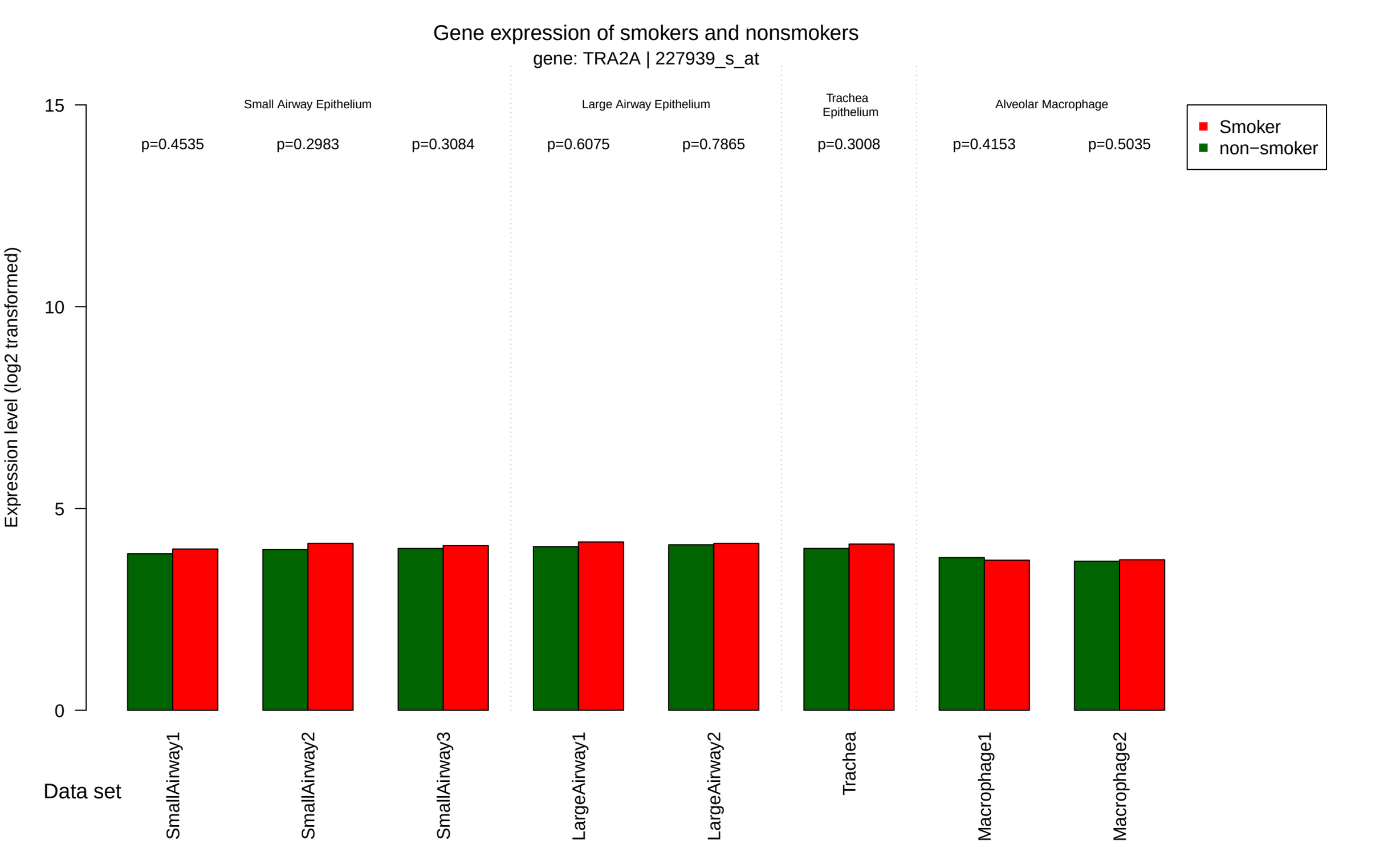
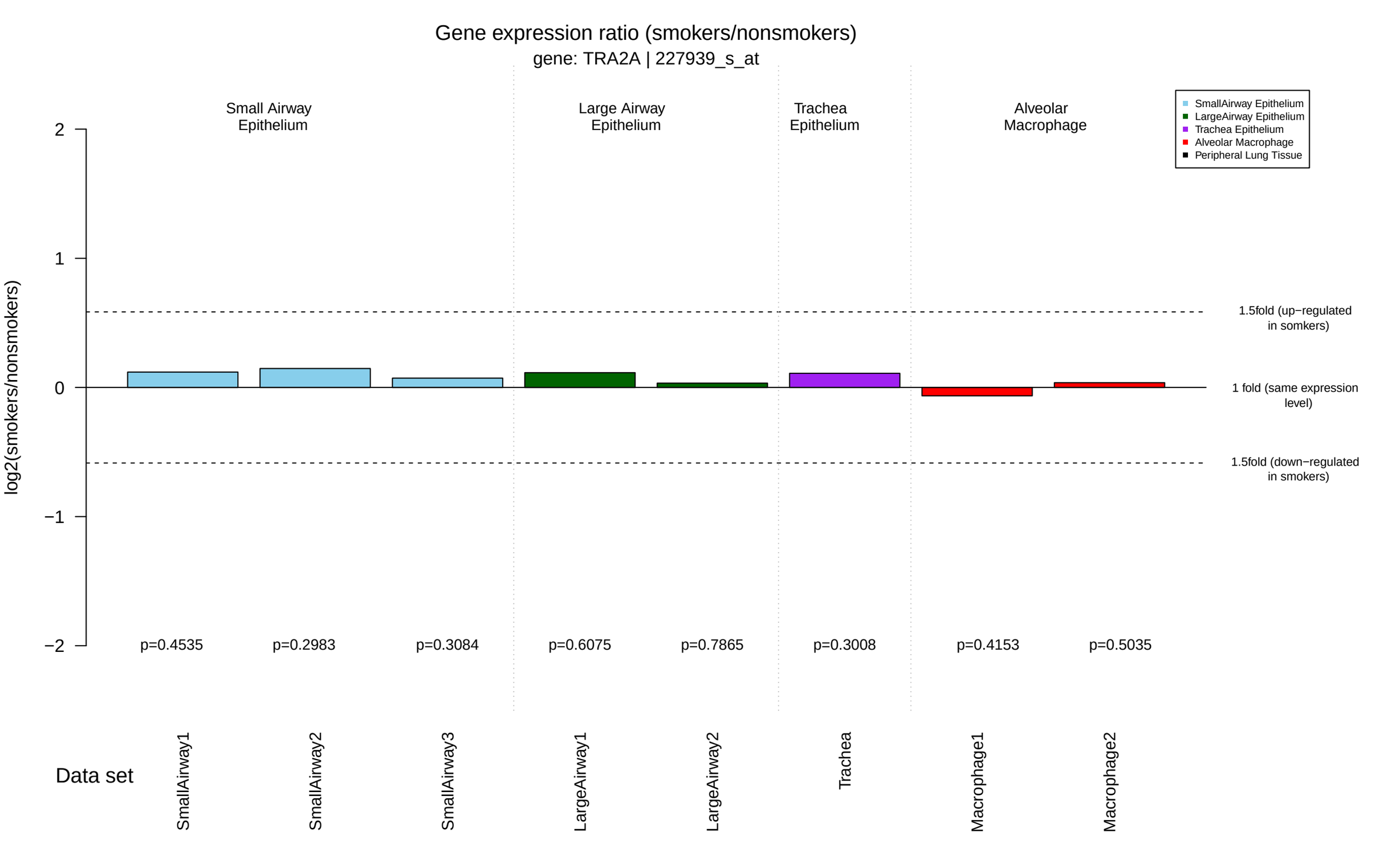
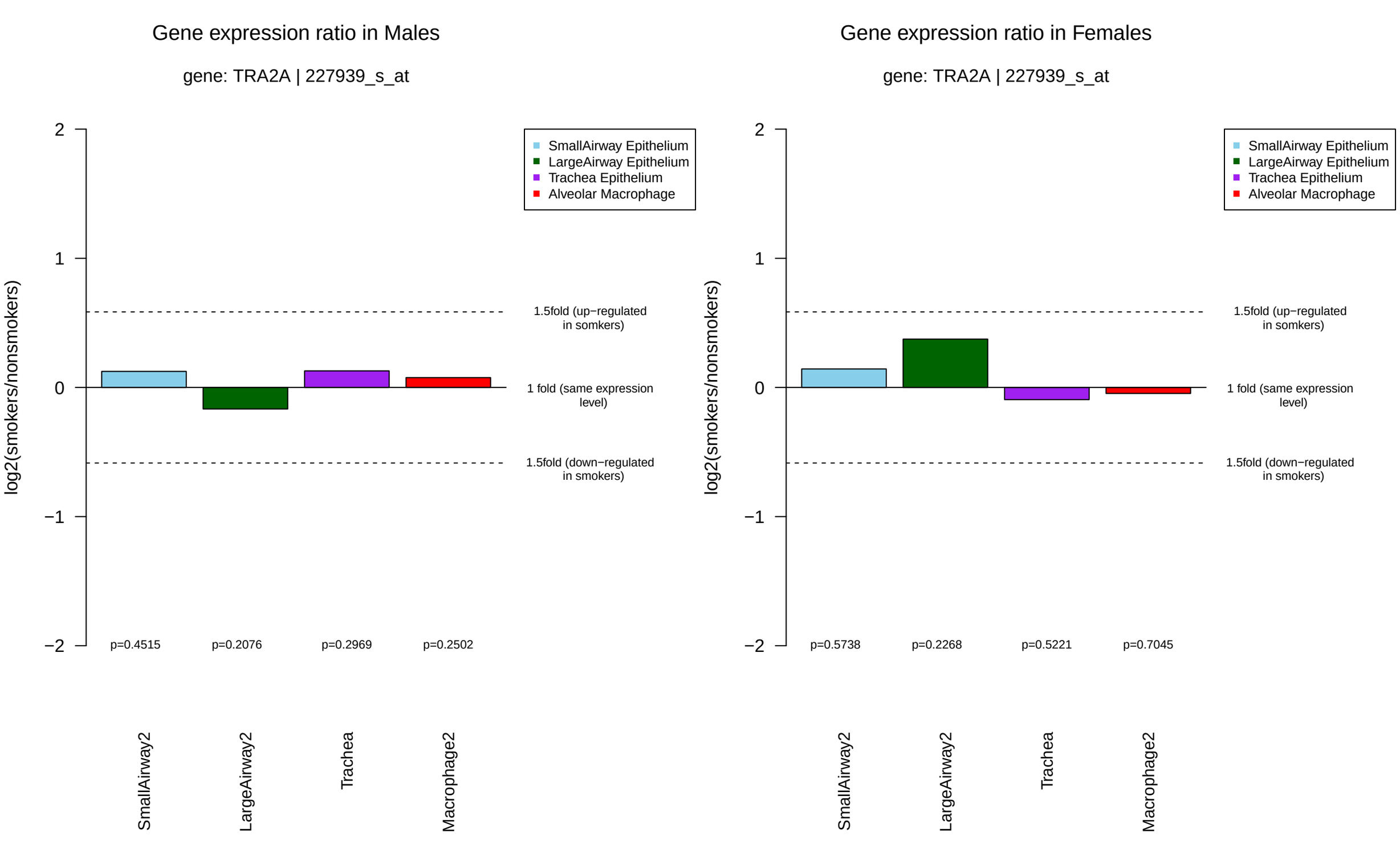
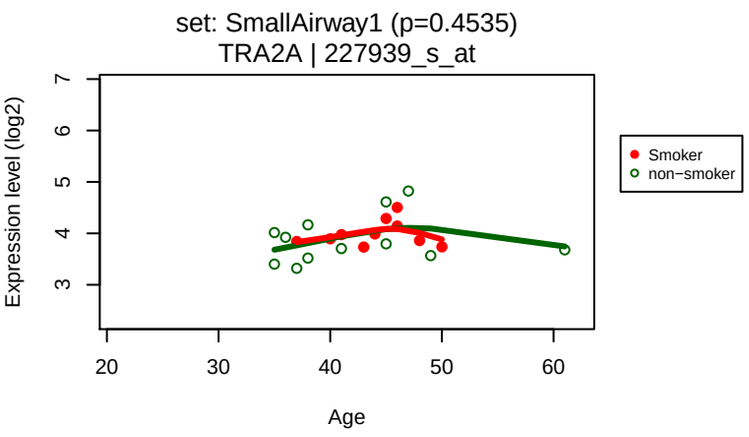
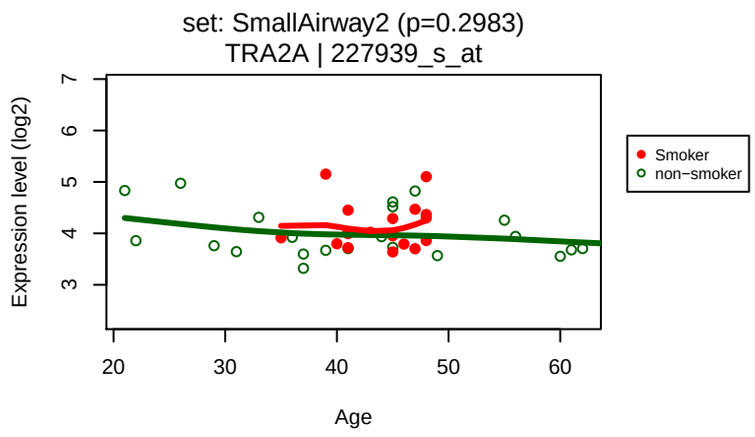
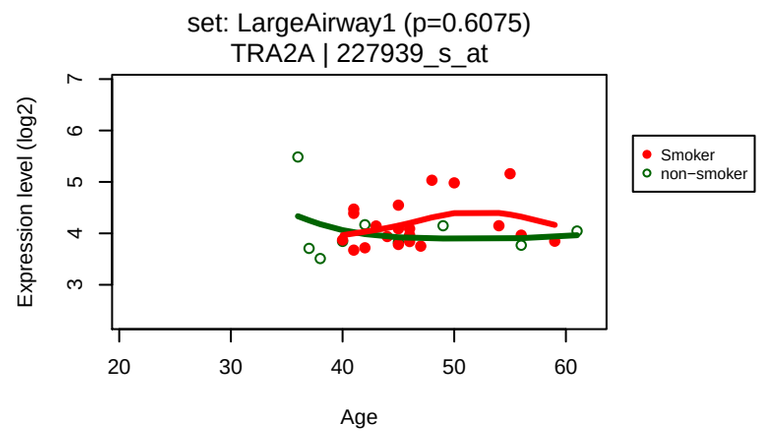

Gene expression of smokers and nonsmokers
gene: TRA2A | 227939_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4535
p=0.2983
p=0.3084
p=0.6075
p=0.7865
p=0.3008
p=0.4153
p=0.5035
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

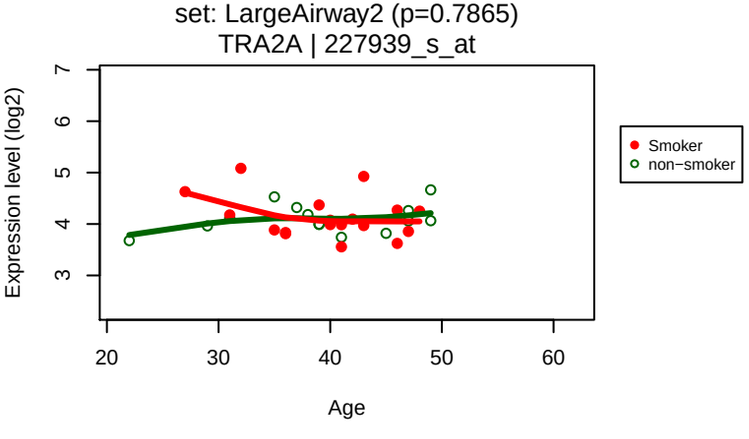

Gene expression ratio (smokers/nonsmokers)
gene: TRA2A | 227939_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4535
p=0.2983
p=0.3084
p=0.6075
p=0.7865
p=0.3008
p=0.4153
p=0.5035
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

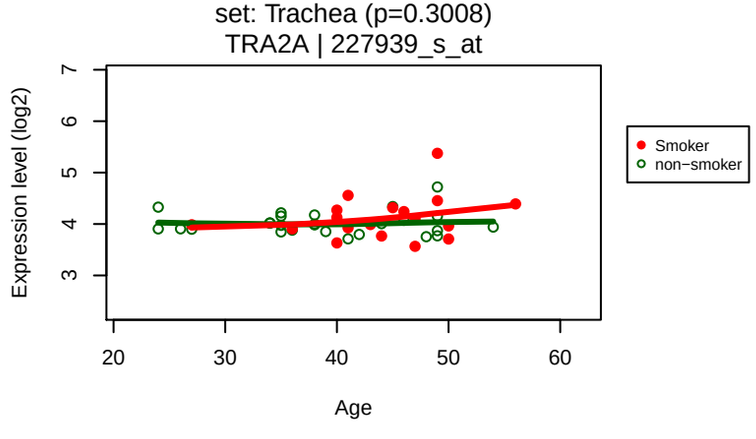

Gene expression ratio in Males
gene: TRA2A | 227939_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4515
p=0.2076
p=0.2969
p=0.2502
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TRA2A | 227939_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5738
p=0.2268
p=0.5221
p=0.7045
SmallAirway2
LargeAirway2
Trachea
Macrophage2

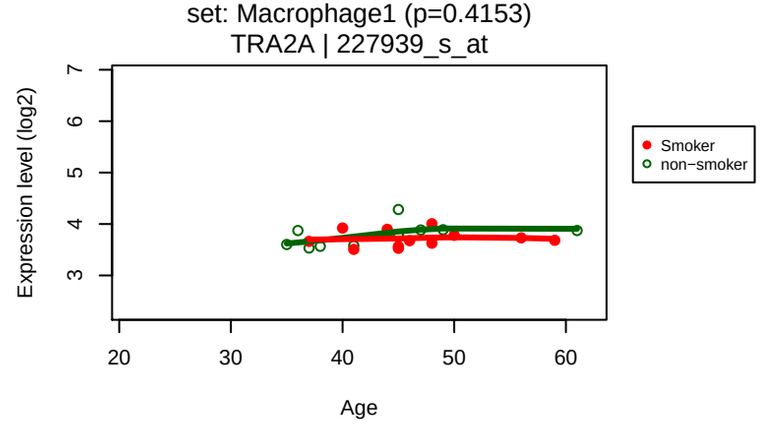

set: SmallAirway1 (p=0.4535)
TRA2A | 227939_s_at
Expression level (log2)
Age
Smoker
non−smoker

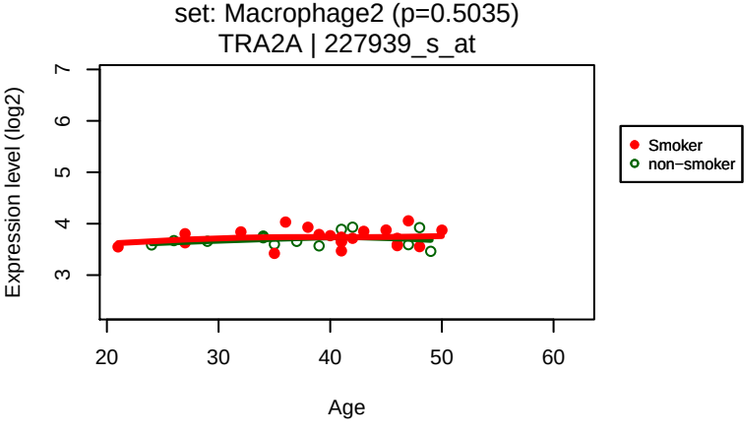

set: SmallAirway2 (p=0.2983)
TRA2A | 227939_s_at
Expression level (log2)
Age
Smoker
non−smoker

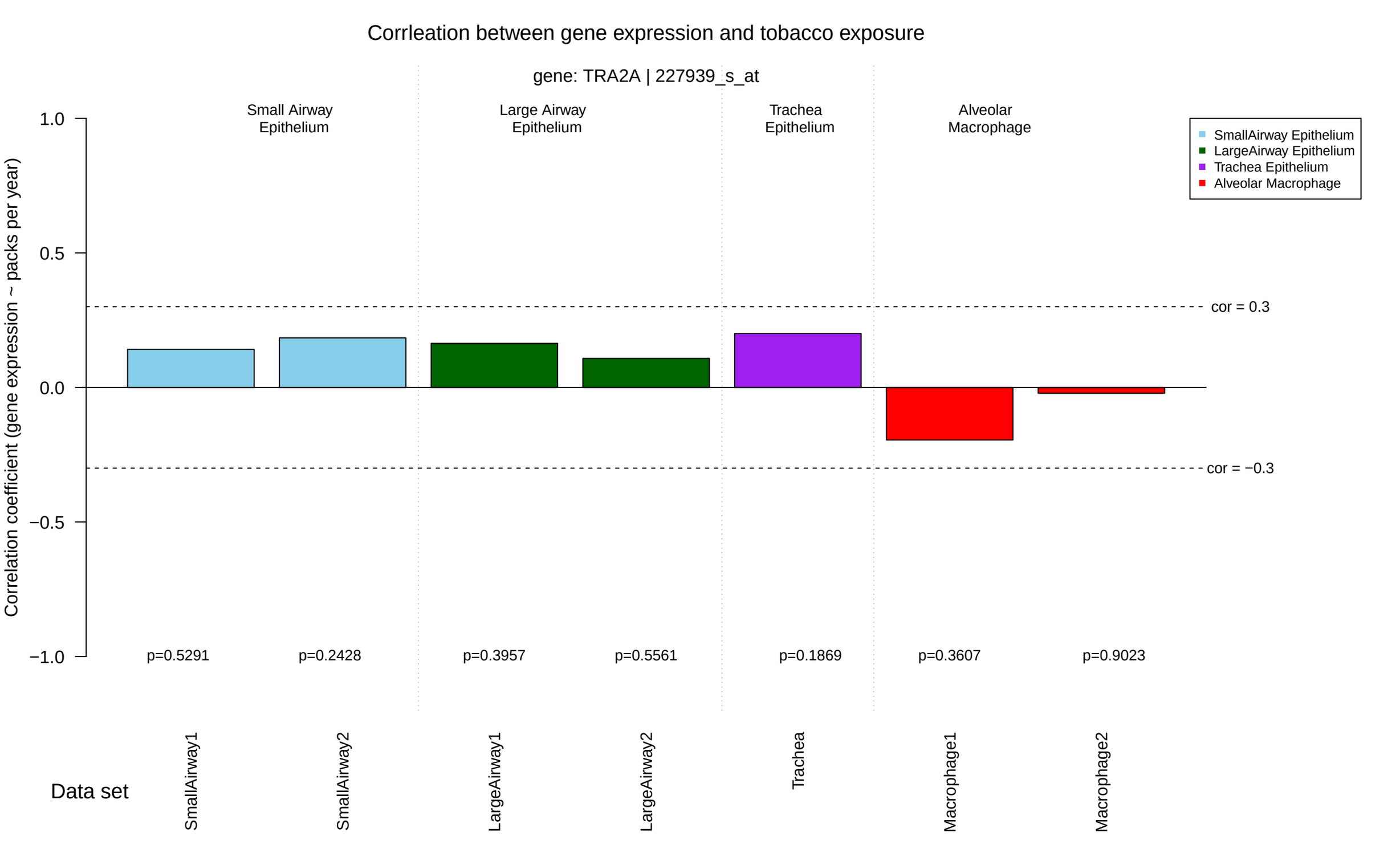

set: LargeAirway1 (p=0.6075)
TRA2A | 227939_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7865)
TRA2A | 227939_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3008)
TRA2A | 227939_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4153)
TRA2A | 227939_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5035)
TRA2A | 227939_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TRA2A | 227939_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5291
p=0.2428
p=0.3957
p=0.5561
p=0.1869
p=0.3607
p=0.9023
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2